



D.Y. Patil Academic Education Excellence Federation's

Dr. D. Y. Patil Technical Campus

Polytechnic | Engineering | MCA

(Approved by AICTE, Recognized by DTE, Govt. of Maharashtra, Affiliated to Savitribai Phule Pune University, Pune)

Sr.No.32/1/A/7, Near Talegaon Railway Station, A/P Varale, Talegaon Dabhade, Tal-Maval, Dist-Pune 410 507

Website: www.dypatiltes.com Email: principal@dypatiltes.com

AICTE ID: 1- 43358498993 | SPPU PUN CODE: CMMP021930 | ISHE CODE: C -71504 | DTE CODE: 6991

Dr. D. Y. Patil

Founder Dr. D. Y. Patil Group

Dr. Sushant V. Patil

Chairman

Adv. Anuja Sushant Patil

Director

Dr. Suresh Shirbahadurkar

Principal

DEPARTMENT OF ELECTRONICS AND COMPUTER ENGINEERING

TOPIC: Microprocessor and Microcontroller(8051)

Prepared By: Mule Gitanjali Satish

Microprocessor and Microcontroller (8051)

1. Introduction to Microprocessor and Microcontroller

A microprocessor executes instructions stored in memory and controls the operation of a digital system.

Examples include 8085 and 8086.

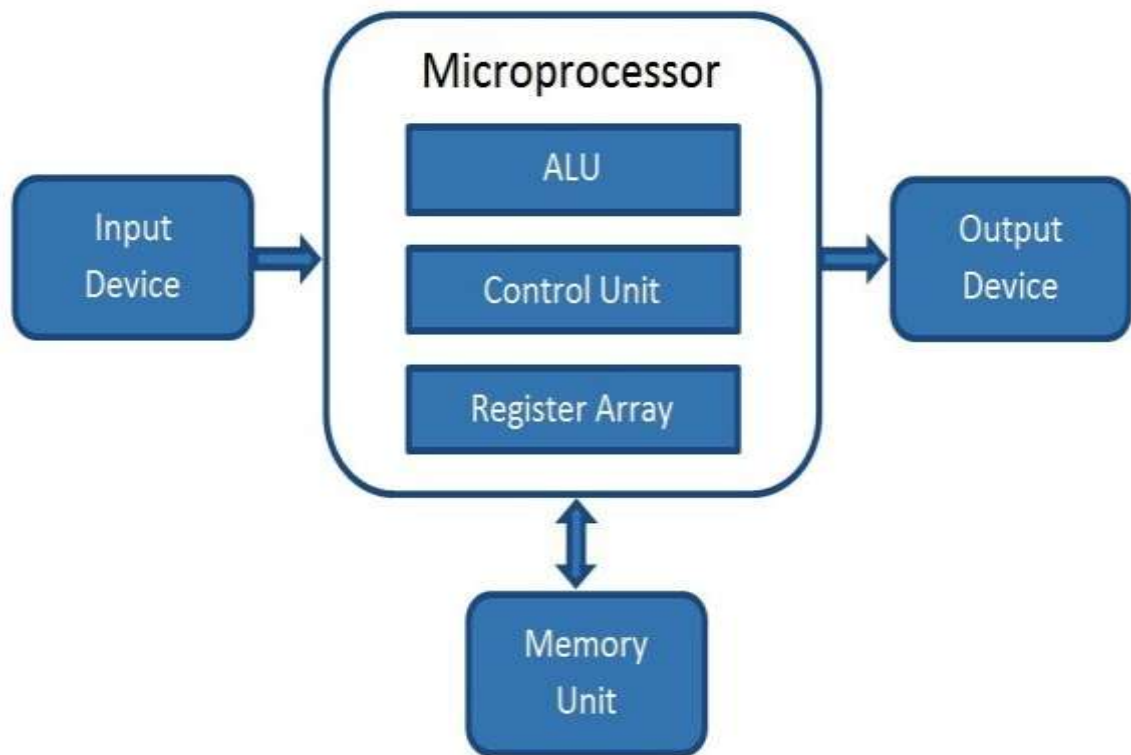
Main components: ALU, Control Unit, Registers and buses.

Applications: computers, general-purpose processing systems, and systems requiring high processing flexibility.

2. Advantages of Microprocessor

- High processing speed
- Performs complex calculations
- Flexible and programmable
- Supports large memory
- Suitable for multitasking
- Easy to upgrade
- Used in computers and advanced systems
- Reliable and compact

3. Block Diagram of Microprocessor



4. What Is a Microcontroller?

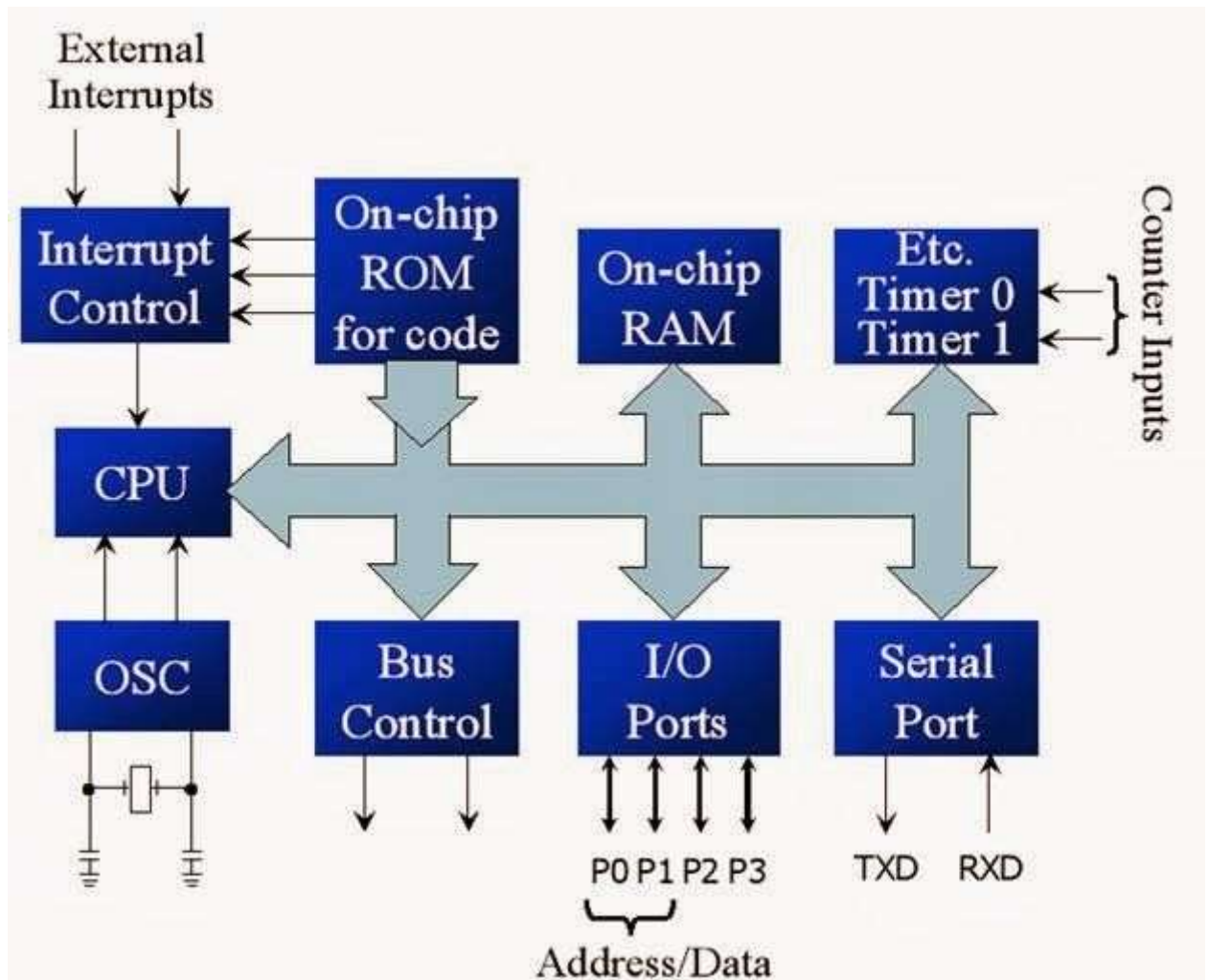
A microcontroller integrates processing, memory and control peripherals into one IC. It is mainly used in embedded systems where a dedicated task must be controlled. Main components: CPU, RAM, ROM/Flash, I/O ports, timers/counters, interrupts and serial communication.

Applications: automation, appliances, traffic control, sensor systems, robotics and IoT devices.

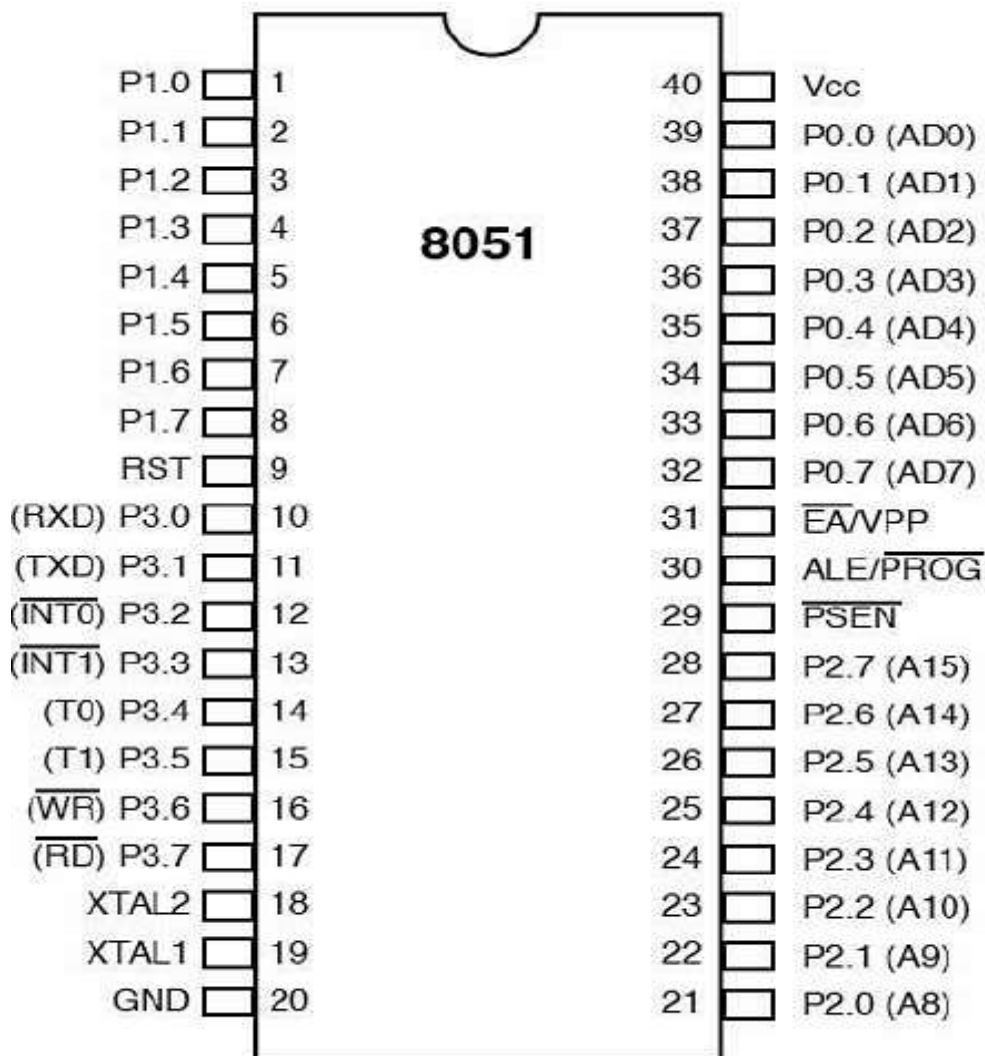
5. Advantages of Microcontroller

- Low cost
- Low power consumption
- Compact size
- Built-in memory and peripherals
- Requires fewer external components
- Easy to program
- Fast response for control applications
- Reliable and suitable for embedded systems

6. Block Diagram of Microcontroller



7. Pin Diagram of 8051



- 8051 has 40 pins.
- Port 0: I/O and address/data bus.
- Port 1: General-purpose I/O.
- Port 2: I/O and address bus.
- Port 3: I/O with special functions.
- VCC & GND: Power supply.
- RST: Resets the microcontroller.
- XTAL1 & XTAL2: Provide clock signal.

- EA, ALE & PSEN: External memory control.

- Instruction Set

- An instruction set is a collection of commands understood by a processor.

- Instructions perform operations such as addition, subtraction and data transfer.

- Microprocessors often support complex instruction sets.

- Microcontrollers use instruction sets optimized for control applications

- Registers

- Registers are small, high-speed storage locations inside the processor.

- They temporarily store data, addresses and instructions.

- Common registers include the accumulator, program counter and status register.

- Registers help improve processing speed.

- Input and Output Ports

- I/O ports allow the controller to communicate with external devices.

- Input devices include sensors, switches and keyboards.

- Output devices include LEDs, motors, displays and relays.

- Ports can often be configured as input or output.

8. Working 8051

- The clock provides timing.

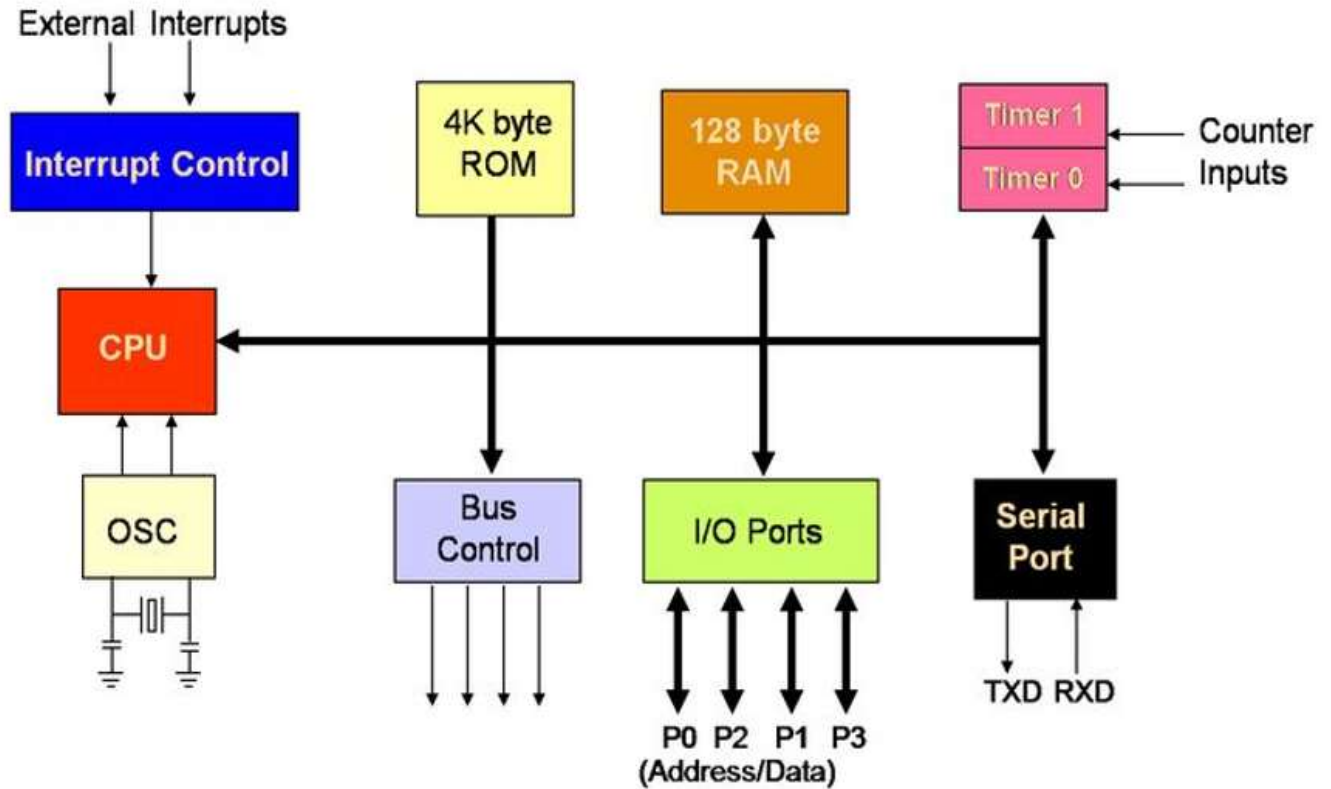
- CPU fetches instructions from memory.

- Instructions are decoded and executed.

- Data is processed and the result is given through output ports.

Flow: Input → Processing → Output

9. 8051 Block Diagram



10. Main Features of Classic 8051

- 8-bit CPU+
- 4 KB on-chip program ROM
- 128 bytes internal RAM
- 32 programmable I/O lines
- Two 16-bit timers/counters
- Full-duplex serial port
- Interrupt system
- Oscillator/clock circuitry

11. Microprocessor vs Microcontroller

Microprocessor vs Microcontroller

Memory	Needs external memory & storage, flexible	On-chip memory (RAM, ROM, EEPROM)
Peripheral 	Need additional part via external bus, ideal for complex, powerful applications.	Integrated peripheral SPI, UART, RTI, ADC for direct HW control
Clock Speed 	Fast GHz for heavy duty tasks 	KHz - MHz, efficient for specific tasks. 
Power Consumption 	High power consumption, no low power option. 	Low power consumption with power saving options 
Connectivity	Handles high-speed transfer	supports low to moderate speed communication I2C, SPI, UART
Operating System 	Need full-fledge OS (Linux, Windows)	OS is optional (no-OS or light weight RTOS)
Usecase	For generic computing, high complexity capacity system	For compact systems, battery powered, or logic processing devices
Cost	Higher cost — suited for performance-driven applications	Lower cost — economical for mass production.

12.Applications of 8051

- Traffic light control
- Digital clock and counters
- Temperature monitoring
- Motor control
- Security systems
- LCD and keypad interfacing
- Industrial automation
- Basic IoT and sensor-control projects

13.Advantages

- Simple and easy to learn
- Compact embedded solution
- Built-in timers and serial communication
- Multiple programmable I/O pins
- Suitable for basic control applications

14.Future Trends

- Artificial intelligence at the edge
- Internet of Things
- Smart sensors
- Autonomous vehicles
- Wearable electronics
- Energy-efficient processors

15.Limitations

- Limited Memory – Classic 8051 has only 4 KB ROM and 128 bytes RAM, which is small for complex applications.
- 8-bit Processor – It processes 8-bit data, so it is slower for large or complex calculations compared with modern 16-bit and 32-bit microcontrollers.
- Limited Processing Speed – The original 8051 operates at relatively low clock speeds compared with modern microcontrollers.
- Limited Built-in Peripherals – The classic 8051 has fewer built-in features such as ADC, DAC, and advanced communication interfaces.
- External Components Required – Some applications need external memory, ADC, drivers, or other hardware, increasing circuit complexity.